Numerical Analysis Assignment 7

刘金穗 - 2230018044

Question 1

(a) the best line is $p_1(x) = 1.0714x + 0.8571$.

(b) The best parabola $p_2(x) = -0.1657x^2 + 1.9475x + 0.3481$.

(c) The best degree 3 polynomial is $p_3(x) = 0.3500x^3 - 2.5500x^2 + 5.2000x$. The degree 3 interpolating polynomial is $p_3(x) = 0.3500x^3 - 2.5500x^2 + 5.2000x$. It is the same as the best degree 3 polynomial through the data points.

Question 2

(a) The degree 2 interpolating polynomial is

$$p_2(x) = -\frac{4}{\pi^2}x\left(x - \frac{\pi}{2}\right) + \frac{2}{\pi}x$$

(b) The approximation for $\sin(\pi/4)$ is $p_2(\pi/4) = 0.75$

(c) The error is

$$|e(x)| = |f(x) - p(x)| \leq \left|\frac{\sin^{(3)}(0)}{3!}\frac{\pi}{4}\left(\frac{\pi}{4} - \frac{\pi}{2}\right)\left(\frac{\pi}{4} - \pi\right)\right| = 0.2422$$

(d) The exact error is $\left|f\left(\frac{\pi}{4}\right) - p\left(\frac{\pi}{4}\right)\right| = 0.0429$.

Question 3

(a) Denote that

$$f(x) = \sum_{j=0}^{n} x_j^k L_j(x) - x^k$$

Then the degree of $f(x)$ is at most n, which is the equation $f(x) = 0$ have at most n roots. Assume that $f(x) \not\equiv 0$, but for $i = 0, 1, \ldots, n$, then

$$f(x_i) = \sum_{j=0}^{n} x_j^k L_j(x_i) - x_i^k = x_i^k - x_i^k = 0.$$

Then, $n + 1$ zeros of $f(x)$ is found, which gives a contradictory. Therefore,

$$\sum_{j=0}^{n} x_j^k L_j(x) \equiv x^k$$

(b) Since for $j = 0, 1, \ldots, n$

$$(x - x_j)^k = x_j^k + \binom{n}{1} x_j^{n-1}(-1)x + \cdots + (-1)^k x^k.$$

Then,

$$\begin{aligned}\sum_{j=0}^{n}(x - x_j)^k L_j(x) =& \sum_{j=0}^{n} x_j^k L_j(x) + (-x)\binom{n}{1}\sum_{j=0}^{n} x_j^{k-1} L_j(x) + \cdots + (-x)^k \sum_{j=0}^{n} L_j(x) \\ =& \left[\binom{k}{0} - \binom{k}{1} + \binom{k}{2} + \cdots + (-1)^k \binom{k}{k}\right] x^k\end{aligned}$$

If k is odd, by $\binom{n}{k} = \binom{n}{n-k}$,

$$\sum_{j=0}^{k}(-1)^j \binom{k}{j} = \binom{k}{0} - \binom{k}{k} + \cdots + \binom{k}{\frac{k-1}{2}} - \binom{k}{\frac{k+1}{2}} = 0.$$

If k is even, because that $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k+1}$, then

$$\begin{aligned}\sum_{j=0}^{k}(-1)^j \binom{k}{j} =& \binom{k-1}{0} - \left[\binom{k-1}{1} + \binom{k-1}{2}\right] + \cdots \\ &+ \left[\binom{k-1}{k-2} + \binom{k-1}{k-1}\right] - \binom{k-1}{k-1} \\ =& 0.\end{aligned}$$

Therefore

$$\sum_{j=0}^{n}(x - x_j)^k L_j(x) = \left[\binom{k}{0} - \binom{k}{1} + \binom{k}{2} + \cdots + (-1)^k \binom{k}{k}\right] x^k = 0.$$